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Using the Body Weight Forward Lunge to Screen an Athlete's Lunge Pattern

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SUMMARY

THE LUNGE PATTERN IS CONSIDERED A FUNDAMENTAL MOVEMENT PATTERN DUE TO ITS OCCURRENCE IN ACTIVITIES OF DAILY LIVING AND SPORT-SPECIFIC TRAINING. IT IS A PATTERN OFTEN USED BY SPORT MEDICINE PROFESSIONALS AND STRENGTH AND CONDITIONING SPECIALISTS TO IMPROVE PROPRIOCEPTION AND STRENGTH. THE FORWARD LUNGE HAS BEEN ESTABLISHED AS AN EFFECTIVE EXERCISE FOR IMPROVING COORDINATION AND MUSCLE BALANCE TO ENHANCE PERFORMANCE IN UNILATERAL ACTIVITIES SUCH AS SPRINTING AND CHANGE OF DIRECTION. DUE TO THE PREVALENCE OF THE LUNGE PATTERN IN SPORT TRAINING, SCREENING AN ATHLETE'S LUNGE PATTERN BEFORE EXTENSIVE LOADING OF MUSCLE OCCURS MAY BE A BEST PRACTICE.

INTRODUCTION

The lunge pattern is 1 of 7 fundamental movement patterns performed in activities of daily living, sport, and sport-specific training (11). The 7 fundamental movement patterns are the squat pattern, lunge pattern, upper-body push pattern, upper-body pull pattern, bending pattern, twisting pattern, and gait pattern (11). This article focuses on the lunge pattern given its popularity to sport and sport-specific training.

The forward lunge, the most popular lunge pattern exercise, exaggerates the movement that occurs in the lower body during the gait cycle (14). The forward lunge involves: a) calcaneal eversion; b) talar plantar flexion and adduction; c) tibial internal rotation; d) knee flexion, extension, and abduction; and e) hip flexion, extension, and adduction (14). Various lunge pattern exercises have been used as assessment tools for measuring strength, flexibility, and balance (13,14,36). Hybrids of the forward lunge have been used to screen the functional movement of the lower extremities (13). Given the relevance of the lunge pattern to sport and the necessity of the strength and conditioning specialist to load the movement pattern to enhance performance, screening an athlete's lunge pattern may have benefits. It is widely accepted that an important responsibility of strength and conditioning specialists is to reduce the incident of injury.

Understanding how an athlete moves is of critical importance if strength training is to be a performance enhancer rather than an injury mechanism. Identifying faulty strategies of movement and correcting those strategies prior to prescribing heavy loads and advanced loading schemes may ensure the athlete's long-term development. This review aims to establish the forward lunge as an effective screen of the lunge pattern. Existing empirical evidence highlighting the regions of the body that have been identified to be areas most susceptible to breakdowns in technique is presented. Biomechanical rationale is provided to assist the reader in understanding the consequences poor forward lunge technique has on movement competency, the incidentce of injury, and sport performance.

KINEMATICS AND KINETICS OF THE FORWARD LUNGE

A forward lunge can in the simplest terms be described as an elongated forward step, flexing the lead hip and knee and dorsiflexing the lead ankle while lowering the body toward the floor (20,22,23). To screen a lunge pattern, the strength and conditioning professional should understand how the body is designed to move and the consequences to movement efficiency, force production, and injury occurrence if function is compromised. The variables that may affect the ability of

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assessment; functional; movement; strength programming; injury

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Table 1
Optimal forward lunge criteria and viewing position

Anatomical region	Optimal viewing position	Cook (13)	Brandon (9)	Graham (22,23)	Summary	Poor pattern	Possible contributors to poor pattern
Head	Front and side	Centrally positioned	Central over feet, neutral position	Erect facing forward	Straight and centrally aligned	Positioned forward or tilted to one side	Muscle imbalance of the neck flexors and extensors
Thoracic spine	Side	Straight	Vertical, shoulder above hips	Erect, shoulders above hips	Straight or slightly extended	Flexed	Poor or imbalanced trunk extensors
Lumbar spine	Side	Neutral	Neutral, no back extension to assist hip extension, remains tall and stable	Erect	Neutral	Flexed or excessively extended	Weak deep abdominals and gluteus maximus and/or poor coordination, poor flexibility in hip flexors
Hips	Front	Horizontally aligned	NA	Aligned with the lead knee and ankle	Horizontally aligned	Dropping to one side, rotated medially or laterally	Weak deep abdominals and gluteus maximus, medius, and minimus
Knees	Front and side	Aligned with the hip and foot	Front knee points forward over the small toes and is above the ankle, back knee flexed	Front knee over the lead ankle	Front knee in line with the small toes and over the lead ankle	Movement medially or laterally during the lunge	Poor sequencing of gluteus maximus, medius, and minimus and deep core muscles. Restricted range of motion with hip flexors, quadriceps
Ankles	Front and side	Front—aligned with the knee and hip. Side—directly under the knee	Aligned with the knee	Aligned with the knee	Aligned with the knee	Limited dorsiflexion	Restricted mobility due to stiffness or restricted range of motion of the gastrocnemius and soleus muscles
Feet	Front and side	Front foot flat, back foot positioned on the toes aligned with the heel aligned with the knee	Front foot flat, back foot on the ball of the foot	Front foot flat, back foot on the ball of the foot	Front foot flat, back foot on the ball of the foot with toes flexed. Both feet aligned and balanced	Rolling in or out of the lead foot	Poor stability and proprioceptive response

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an athlete to complete a proper lunge pattern have been acknowledged to be anthropometrics, handedness, previous injury, lack of coordination, range of motion, and balance (2,13,20,26,35). Table 1 summarizes published criteria detailing what is considered to be correct forward lunge technique. Figure 1 illustrates a forward lunge demonstrating good alignment, mobility, stability, and balance. The sections below discuss the kinematic and kinetic characteristics of the lunge pattern, with specific focus on those regions of the body that directly influence forward lunge movement competency.

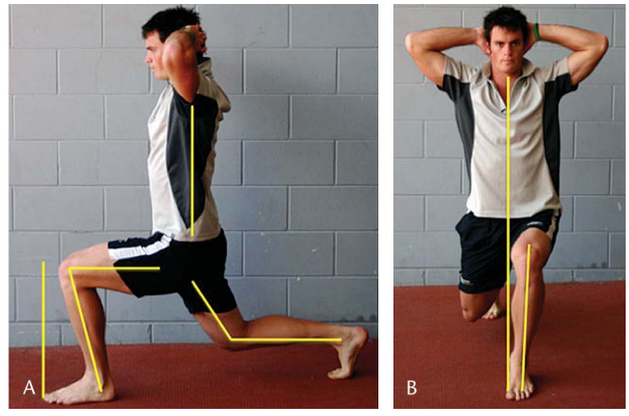


Figure 1. Optimal forward lunge technique, viewed from the side (A) and the front (B).

ANKLE MOBILITY

The ankle joint complex consists of 3 joints, namely, the ankle joint, subtalar joint, and the midtarsal joint (30,38). The motions that take place at the ankle are dorsiflexion/plantarflexion, inversion/eversion, and axial rotation (38). Given the ankle's range of motion capability in frontal, sagittal, and transverse planes and because none of the motions take place exclusively at one joint, the ankle has been identified as a mobility joint (13,35,38). During the performance of the forward lunge, ankle mobility on both the lead and trail legs is critical to ensure a balanced biomechanically correct lunge pattern (6,13). The inability to control foot position and the lack of ankle mobility have been reported to encourage movement strategies that have been identified to be injury mechanisms (33,35).

Movement strategies such as turning out of the feet, turning in of the feet, dropping of the arch, and/or lifting of the lead heel off the ground are considered faulty during the performance of a lunge pattern (4,16–18,20,32,35). The effects of faulty strategies on the body during sport and sport-specific training are not entirely understood. Kovacs et al. (32) and Flanagan et al. (20) found that various biomechanically specious positions of the ankle and foot during activities of daily living (i.e., squatting and lunging) resulted in higher forces incurred at the knee and hip. It is believed that increases in joint forces resulting from bad technique during movement overstress joint ligaments and tendons (6,7,20,40). An indicator of good ankle mobility during a forward lunge is when the athlete can maintain a flat foot position of the lead leg and an aligned flexed foot position during full hip, knee, and ankle flexion (Figure 1). An athlete must have good ankle mobility to perform a good lunge pattern (13).

KNEE CONTROL

During a forward lunge, the knee of the front and back leg should be aligned with the hip and ankle during flexion and extension (5,8,13,31). The knee joint is the largest joint in the body and

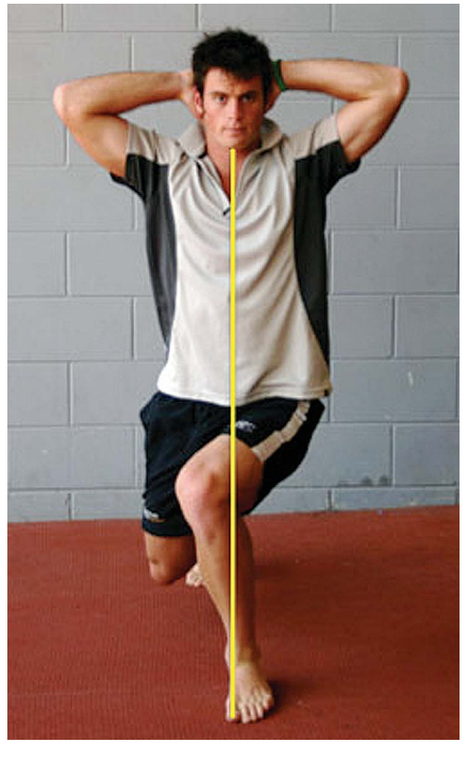


Figure 2. Medial knee motion, on the forward lunge, which is considered improper technique.

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is a modified hinge joint made up of the tibiofemoral and patellofemoral joints, which enable flexion in a posterior direction and extension in the anterior direction (30). The knee joint is not designed to accommodate excessive mediolateral or anteroposterior movement (Figure 2) (16–19). The cause of mediolateral movement of the lead knee during a forward lunge is hypothesized to be poor strength or activation of the rectus femoris, hamstrings, and hip abductor and adductor muscles (12,35).

The hamstrings and the rectus femoris and the gastrocnemius that attach to the knee and ankle disadvantage the knee if they are weak or fail to activate at the right time and may contribute to a poor movement pattern (13,35). In addition, when observing the forward lunge from the side, the athlete should appear to have stepped out far enough so that the lead knee is directly over the lead foot and the heel remains in contact with the ground as the athlete's center of mass is observed to be moving toward the ground (23).

When the center of mass appears to be moving more forward than down and the heel of the front foot raises from the ground to accommodate the forward momentum, there is less emphasis on challenging hip mobility and an increase in patellofemoral shear force has been reported (Figure 3) (3). In addition, athletes using the aforementioned movement strategy often report knee pain. There are many variables that have been identified as contributors to an athlete's inability to control knee alignment: weak or poor activation of the gluteus muscles, over- or underdeveloped quadriceps muscles, and poor mobility in the hips and ankles (3,6,9,14,27,36).

Researchers appear conflicted about the causes of knee injuries related to an athlete's inability to control knee alignment during movement (16,18,21, 28,29,37,41). Nonetheless, it is clear that mediolateral movement of the knee during the eccentric and concentric phases of a lower limb exercise is contraindicated. Therefore, a primary benefit of a movement screen is that the strength and conditioning specialist can identify that an athlete has difficulty controlling knee alignment during movement and can recommend

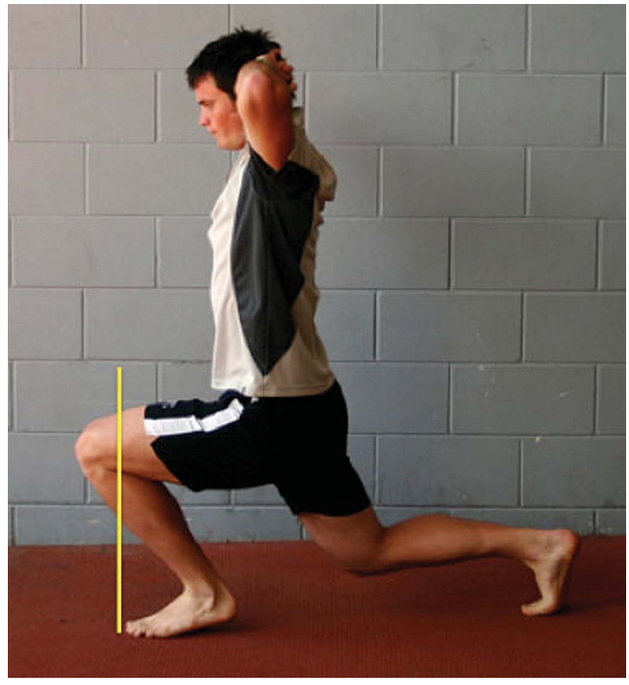


Figure 3. Knee position in front of the toes on decent, which is considered improper technique.

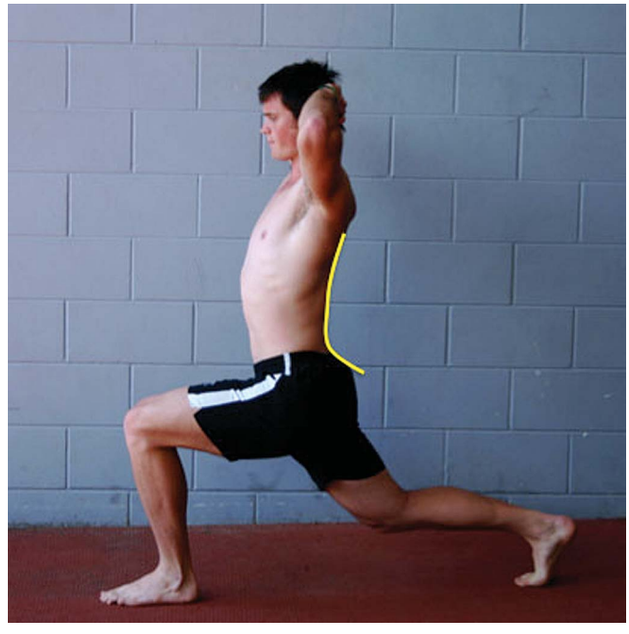


Figure 4. Lumbar extension providing greater hip mobility and stride length. Excessive lumbar extension is considered improper technique.

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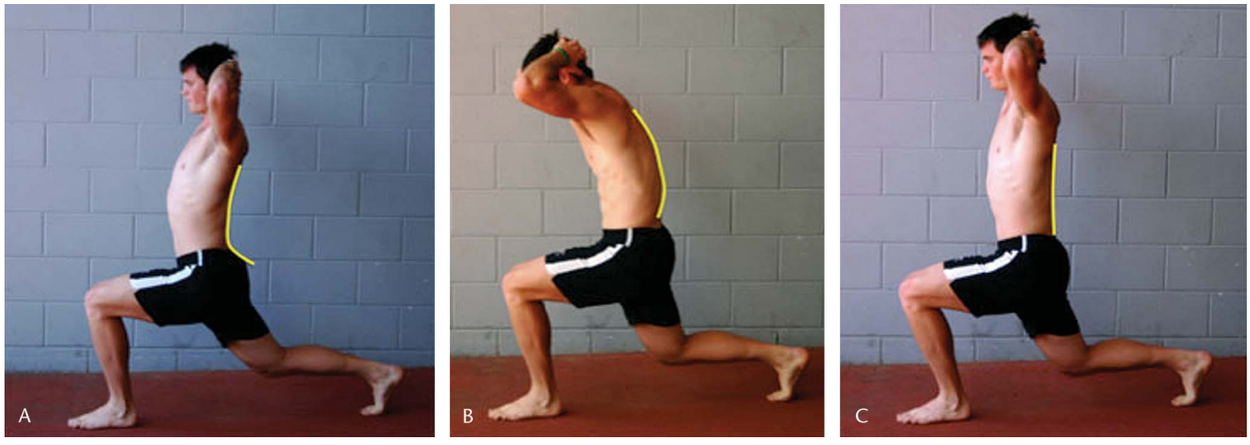


Figure 5. (A) Extended, (B) flexed, and (C) neutral lumbar spine during the lunge. Note that spinal flexion is considered improper technique, and excessive spinal extension is considered improper technique.

a thorough assessment conducted by a sports medicine professional.

HIP MOBILITY

The hip joint is a ball-and-socket joint that is capable of motion in all 3 planes: sagittal (flexion and extension), frontal (abduction and adduction), and transverse (medial and lateral rotation) (24,25). One of the primary roles of the hip joint is to provide a pathway for transmission of forces between the lower extremity and pelvis during activities such as sprinting and change of direction (25). Hip range of motion is considerable with flexion between 0 and 135° and extension 0 and 15° (25).

Table 2
Lunge pattern progressions

Coaching points	Level 1 (assisted)	Level 2 (body weight)	Level 3 (external load)	Level 4 (eccentric)	Level 5 (plyometric)	Interventions
Head: centered	In-place lunge with bands (Figure 6)	Forward lunge	Forward lunge	Drop and stick split lunge (Figure 9)	Split jumps	Alt hip march (Figure 11)
Shoulders/T-spine: held down away from ears/neutral position			Lateral lunge (Figure 7)		Box split jumps	Band pull hip flexion (Figure 12)
Lumbar: neutral			Back lunge (Figure 8)		Scissor jumps (Figure 10)	Crazy Carpet forward lunge (Figure 13)
Hips: horizontally aligned, 90/90 stance						
Knees: aligned with the hips and feet						
Ankles: aligned with the knee						
Feet: heel of lead foot in contact with the floor						

Progression to the next level should only occur if the athlete can demonstrate the exercises detailed in each level with all coaching points maintained through a complete range of motion.

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During a forward lunge, mean hip range of motion has been reported to be 95 ± 27° of flexion (28). Hip range of motion can appear greater if pelvic and lumbar extensions are allowed to take place (Figure 4) (25,28). Forces at the hip during a forward lunge have been reported to be 1.25 and 1.31 times the body weight during the downward and upward phases of the movement (20). Posterior movement of the pelvic and lumbar extensions are movement strategies reported to allow greater hip mobility (25,28,30,31,35). However, when an increase in hip mobility is achieved through pelvic instability and lumbar extension, the forces incurred by those 2 regions have been reported to increase 10% to 30% (20).

When an athlete performs a forward lunge, the hips should remain parallel with the ground (9). There should be no mediolateral rotation or lateral dropping of the hip (9,13). The hips should appear stable to accommodate the mobility necessary to facilitate a good lunge pattern.

TRUNK (LUMBAR STABILITY)

According to researchers, the trunk should remain vertical with the lumbar spine in a neutral position (Figure 5) (22,23). Given the prevalence of low back pain and injuries experienced by athletes with notoriously tight hip flexor muscles, it is critical that the lumbar spine be monitored throughout the lunge pattern. Lunging with an external load and excessive lumbar extension has been reported to dramatically increase compressive forces (33,34,39). A 2° increase in extension from a neutral spine position increased compressive stress within the posterior annulus by an average of 16% compared with maintaining a neutral spine position (39). This is particularly important because researchers have found that athletes hyperextend to a significant degree when lifting heavier (60% and 80% of 1 repetition maximum) loads (1,10,39).

Further investigation demonstrated that the compressive strength of a vertebral body is notably reduced with



Figure 6. In-place lunge with bands showing the start position (A) and the lunge position (B).



Figure 7. Lateral lunge.



Figure 8. Back lunge showing the start position (A) and the lunge position (B).

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Figure 9. Drop and stick split lunge: From the start position(A), drop into a split lunge (B) as fast as you can trying to maintain all coaching points at the lowest point.

movement patterns using a lumbar posture that is not neutral (39). It is therefore suggested that the trunk, in particular the lumbar spine, be observed during low- to moderate-intensity activity prior to the prescription of high-intensity training with heavy loads (33,34,39). The forward lunge provides an opportunity for the trunk and lumbar spine to demonstrate its control and gives the strength and conditioning specialists an opportunity to better understand an athlete's tendency through the lumbar region during movement.

HEAD POSITION

There is no research that has investigated the effects of head position on lunge kinematics and kinetics. The only research found that investigated head position and direction of gaze on movement kinematics involved the bilateral back squat. Donnelly et al. (15) found that when the head position and direction of gaze were directed downward, a significant increase in hip and trunk flexion was observed (15). Movement of the head with a downward direction of gaze during a squat movement increased trunk flexion by up to 4.5° (15). Even though the effects of head position during a forward lunge have yet to be studied, it appears from practical experience that a neutral head position with the direction of gaze directed straight is what the strength and conditioning specialist should expect to see when an athlete performs a forward lunge (22,23).

PROGRESSING LUNGE PATTERN MOVEMENT COMPETENCY

Table 2 provides an example of how an athlete's lunge pattern may be progressed based on concerns identified in the screening process (see also Figures 6–13). The levels detailed in Table 2 progresses along a compendium of intensity guided by movement ability. Progression between levels is determined by the athlete's ability to perform each level's exercise with the coaching points maintained as detailed in Table 2. Level 1 uses strength bands to help attenuate the body weight force to enable the athlete to work through a full range of motion (Figure 6). The strength bands also promote activation of the hip flexors and extensors, which assist with controlling lower limb alignment.

Level 2 is the body weight forward lunge that is used to screen the lunge pattern and serves as an effective method for loading the lunge pattern.



Figure 10. Scissor jumps showing the start lunge position (A), middle airborne position (B), and the end lunge position (C).

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Figure 11. Alternate hip march.

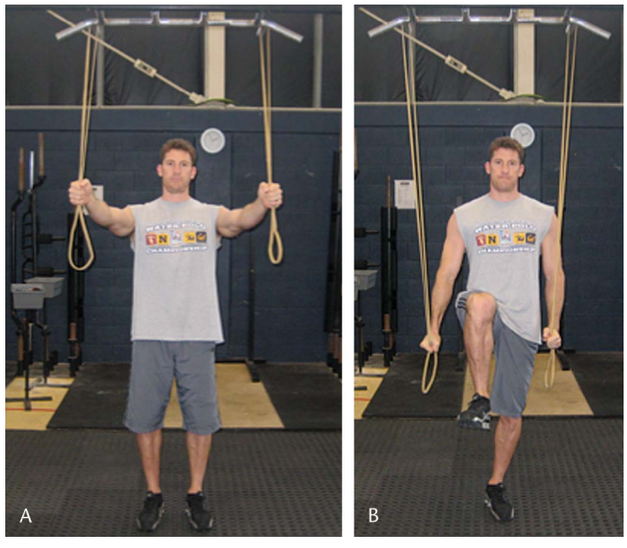


Figure 12. Band pull with hip flexion showing the start position (A) and the hip flexed position (B).

The athlete should demonstrate several body weight forward lunges with minimal effort before progressing to the next level.

Level 3 introduces external loading in the form of free weights, medicine balls, and other isoinertial modalities (Figures 7 and 8). Level 4 focuses on the eccentric phase of the lunge pattern. It provides an opportunity for the athlete to demonstrate control via strength within the lunge pattern range of motion where the most joint forces have been recorded (20). Levels 3 and 4 may be used simultaneously in complex loading schemes to challenge the eccentric phase under high loads.

The exercises in level 5 are examples of traditional lower body plyometric drills that provide an opportunity for the athlete to demonstrate movement competency within the lunge pattern at high velocities. Prescribing level 5 exercises before the athlete has demonstrated movement competency under high force with low velocity is not recommended.

The intervention exercises detailed in Table 2 challenge hip flexion and extension, trunk stability, and lower limb control (Figures 11–13). These exercises can be used early in a plan to complement level 1 exercises or used as movement preparation exercises given how they target the muscles and patterning specific to the lunge pattern.

DISCUSSION

The forward lunge exercise has been presented as a valid screen of an athlete's movement competency related to the lunge pattern. The authors recognize the use of the forward lunge, as a screening tool requires further investigation. For example: Are joint kinematics and kinetics required to accurately screen a forward lunge movement? What is the correlation of full 3-dimensional analysis of a forward lunge pattern to standard 2-dimensional video analysis? When an athlete can perform a body weight forward lunge efficiently and effectively, does screening the pattern under

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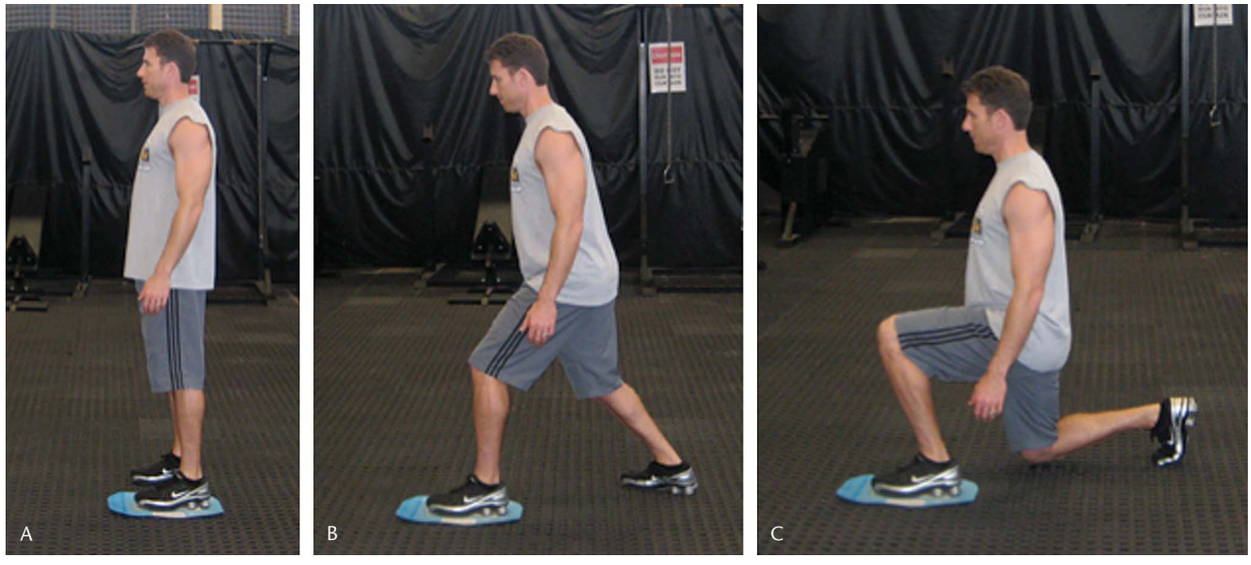


Figure 13. Crazy Carpet forward lunge showing the start position (A), the mid position (B), and the end position (C)—The Crazy Carpet is an 8 × 10 piece of plastic that provides friction resistance to allow a progressive lunge step distance.

high loads and high velocities provide further useful information?

As a screening tool, the body weight forward lunge appears to provide valuable information about an athlete's movement tendency related to lunge pattern exercises. There has been very little research that has investigated the validity of interventions that may be used to correct an athlete's lunge pattern. However, to aid, the strength and conditioning specialists refer to Table 2 for an example of how to progress an athlete's lunge pattern from function to fantastic.

CONCLUSION

The first step toward improving an athlete's movement competency is screening fundamental movement patterns to identify strategies of movement that may contribute to injury and impair performance. The lunge pattern is a fundamental pattern that is common to sport and sport-specific strength training programs. The forward lunge has been proven to be an effective exercise and appears to have prognostic value. To perform a forward lunge correctly, mobility must be present at the ankles and hips and stability must be maintained at the knees and trunk. The athlete may use various movement strategies to accomplish a movement task. However, these strategies may increase the incidence of injury and reduce performance if ignored and excessively loaded. It is recommended that strength and conditioning specialists screen an athlete's lunge pattern with a body weight forward lunge. A simple movement screen performed at the onset of program design may give the strength and conditioning specialists valuable insight into how their athlete moves, further assisting them with developing a strength program that is specific to their needs and capabilities.



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